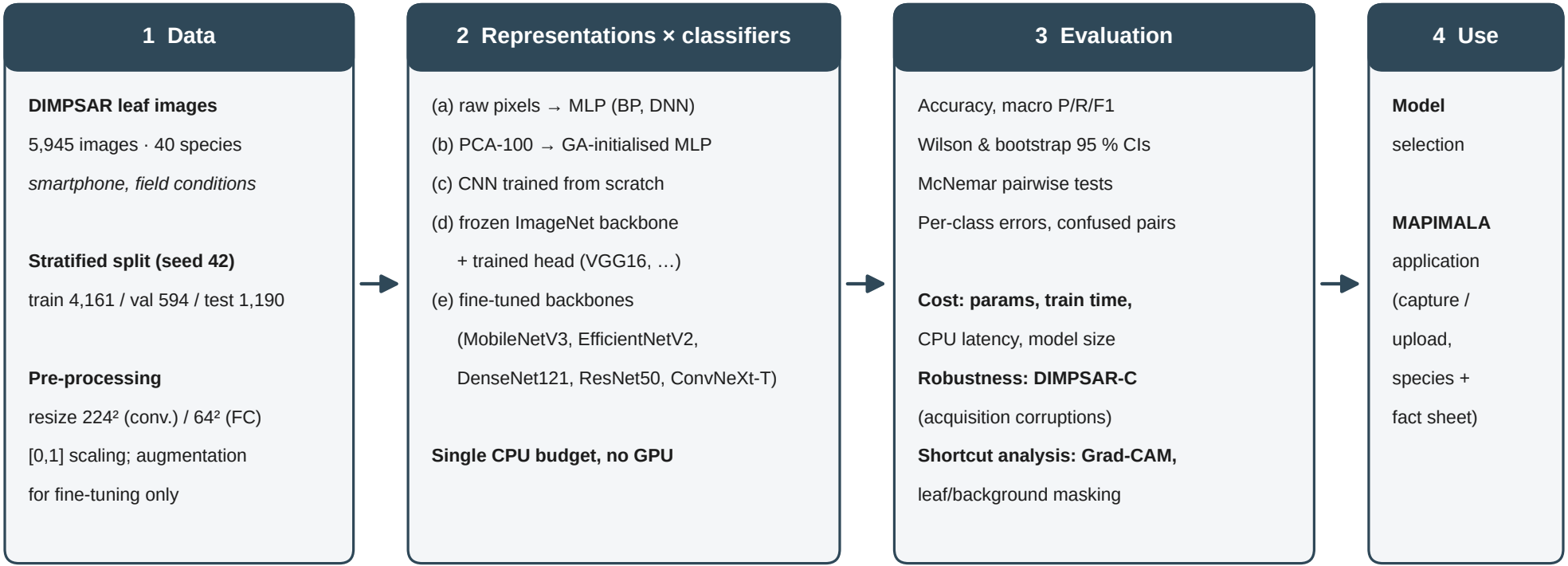
1 Data
DIMPSAR leaf images
5,945 images · 40 species
smartphone, field conditions
Stratified split (seed 42)
train 4,161 / val 594 / test 1,190
Pre-processing
resize 224² (conv.) / 64² (FC)
[0,1] scaling; augmentation
for fine-tuning only
2 Representations × classifiers
(a) raw pixels → MLP (BP, DNN)
(b) PCA-100 → GA-initialised MLP
(c) CNN trained from scratch
(d) frozen ImageNet backbone
+ trained head (VGG16, …)
(e) fine-tuned backbones
(MobileNetV3, EfficientNetV2,
DenseNet121, ResNet50, ConvNeXt-T)
Single CPU budget, no GPU
3 Evaluation
Accuracy, macro P/R/F1
Wilson & bootstrap 95 % CIs
McNemar pairwise tests
Per-class errors, confused pairs
Cost: params, train time,
CPU latency, model size
Robustness: DIMPSAR-C
(acquisition corruptions)
Shortcut analysis: Grad-CAM,
leaf/background masking
4 Use
Model
selection
MAPIMALA
application
(capture /
upload,
species +
fact sheet)